

Calibrating a Bucket

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Created: 18 Feb 2012, last changed 18 Nov 2025

Major revision of software planned: I wish to add the ability to specify uncertainty in any of the measured values; this will cause the printout to show the propagated uncertainty in the length measurement. This should then be used in an example of calibrating a Simplot bucket for 30 lb of strawberries (a 15 L container), which had (in mm) $d = 247.5 \pm 1$, $D = 270(2)$, $L = 285(1)$, and offset $= 7 \pm 0.25$. I marked a 25 mm line on the inside of the bucket to guide the D measurement and used 12 inch Starrett model 39 calipers and a Starrett metric tape measure. The top of the bucket was noticeably elliptical. The bottom was too, but you only notice it by getting the calipers close but undersize, then comparing the diameter at about 6 points around the circumference, gauging the variability by the caliper rattle. In fact, fussy measurements at 6 or 8 points of d and D can be made with rattle measurements and you can estimate the bucket's shape. This would be time-consuming and too much effort for most tasks, yet it would help validate the uncertainty estimates.

This would add another section to this document about the practical nature of calibrating volumes and using the information. In my case, I need to calibrate the output of a water flow meter that measures flow and volume delivered from a garden hose; the use case is to avoid overfilling the water tank of our RV. RV owners are warned of this situation and an overfilled tank can cause damage and expensive repairs.

I have a 5 gallon paint bucket that I'd like to calibrate with marks on the bucket's side to indicate the volume contained in the bucket. I'd like to be able to mark the bucket with different units, such as gallons, liters, and cubic feet. A formula can be derived to do this. We'll give a couple of examples. If you're interested in the derivation, see the appendix. The calibration accuracy will likely be on the order of one to a few percent.

The results given here work for a bucket of any cross-sectional shape and tapering sides:

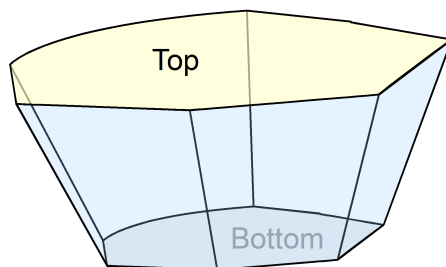


Figure 1

The requirements are that it be a right [frustum](#), which has been misspelled as "frustrum" for millenia. In general, a frustum is a figure between two parallel planes cutting a solid. Here, I'll make the assumption that it's a conical frustum, which means that the lines constructing the sides as shown in Figure 1 all meet at a point. In addition, we'll make the following assumptions:

- ◆ The top and bottom plane figures of the bucket are geometrically similar (they can be polygonal or a non-intersecting curve).
- ◆ The vertical axis is perpendicular to the top and bottom planes and is parallel to the gravitational acceleration vector.
- ◆ The top area is equal to or larger than the bottom area.

5 gallon bucket

The 5 gallon bucket is a common size for paints and industrial chemicals and it's usually a frustum of a nearly circular cone. Since we wind up with one of these buckets after e.g. painting the house, it can be useful to make calibration marks on the exterior or interior to help with various tasks around the house. For mixing concrete, I like to have divisions in cubic feet. For other US-related things, I like to have gallons marked. For general use, having liters marked is handy to allow you to work in SI units.

We'll assume the bucket has a circular cross section. It will taper from the top to the bottom (this is so it can be withdrawn from its mold). We need the following dimensions of the bucket:

D = diameter at inside top

d = diameter at inside bottom

h = height from inside bottom to top

offset = δ = distance from inside bottom to the plane that the bucket sits on

The offset may be a bit confusing; here's a drawing of the bucket's cross section:

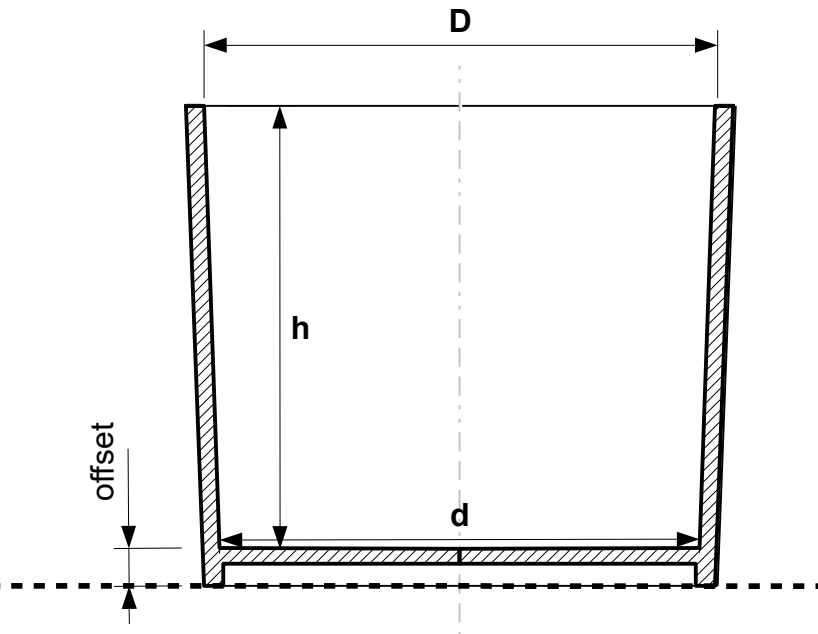


Figure 2

The dashed line is the ground and the offset indicates how much above the ground the inner bucket bottom lies. This is needed because we'll mark the exterior of the bucket using the outside edge that touches the ground as the zero reference (i.e., hook a tape measure to it). I find the easiest way to measure the offset is to put a flashlight in the bottom and illuminate the side; then use a rule to measure from the table top to the darker line of the bottom.

If you want to mark the inside of the bucket with volume calibration marks, set the offset to zero.

A Rheem bucket I have has the following dimensions in inches:

D = 11.34

d = 10.42

h = 14.00

offset = δ = 0.47

We can use equation (1) below to calculate the total volume of the bucket:

$$\text{Bottom surface area} = A_0 = \frac{\pi}{4} d^2 = 85.2757 \text{ in}^2$$

$$\text{Top surface area} = A_1 = \frac{\pi}{4} D^2 = 100.9987 \text{ in}^2$$

Therefore the volume is

$$\begin{aligned} V &= \frac{h}{3} (A_0 + A_1 + \sqrt{A_0 A_1}) \\ &= \frac{14}{3} (85.2757 + 100.9987 + \sqrt{85.2757(100.9987)}) = 1302.37 \text{ in}^3 \end{aligned}$$

Since a gallon is 231 cubic inches, this is 5.64 gallons.

Suppose we want to calibrate the bucket with marks for integer gallons. We use formula (5). First, we need $\sigma = \frac{D}{d} - 1 = 0.08829$. When σ is small, the sides of the bucket aren't tapered a lot; in this case, the formula above (5) in the appendix gives the angle of the side off the vertical as 2.05° .

1 gallon is 231 in^3 ; this is v_x . We can now calculate x from equation (5):

$$x = \frac{1}{\sigma} \left[\sqrt[3]{1 + \frac{3\sigma v_x}{A_0 h}} - 1 \right] = \frac{1}{0.08829} \left[\sqrt[3]{1 + \frac{3(0.08829)231}{85.2757(14)}} - 1 \right] = \frac{0.01680}{0.08829} = 0.1903$$

This x gives the *vertical* distance up the side that the mark needs to be (as a fraction of the height h). To convert this to the slant distance along the bucket's side, we add the offset δ and then divide the result by the cosine of 2.05° :

$$y = \frac{xh + \delta}{\cos(2.05^\circ)} = \frac{0.1903(14) + 0.47}{0.9994} = 3.14$$

Thus, if we hook a tape measure to the outside bottom of the bucket and make a mark 3.14 inches up along the side, this will mark the 1 gallon fill point. The other calibration marks are calculated similarly.

Measuring bucket dimensions

The best tool I have found for making these inside diameter measurements are 12 inch inside calipers (Starrett models 25, 27, 37, 39 or the models 85 or 90 wing dividers with the inside caliper legs). Alternatively, use two pieces of wood like Popsicle sticks clamped together with office binder clamps (usually called measuring sticks). High accuracy isn't required, as the buckets I've measured are often a mm or two out of round -- thus, for best accuracy, you'll want a mean diameter.

What to expect

With good mean diameters of a bucket, you can probably expect your calibration marks to be around one to a few percent accurate (getting 1% accuracy requires care). The first bucket I measured and marked with this method was an old 5 gallon paint bucket and I used a 1000 mL graduated cylinder to check the calibration marks by repeatedly filling the cylinder and pouring it into the bucket.

The graduated cylinder is marked TC/TD 20°C , which means "to contain" and "to deliver" the fluid at 20°C . This Nalgene graduated cylinder came from a friend who had some scrapped laboratory equipment. It is made from clear plastic, is 420 mm tall excluding the base, and has an inside diameter at the top of 58.5 mm. The inside diameter is tapered, as I could feel it with 8 inch Fay calipers. The calibration marks are molded into the side of the cylinder and are every 10 ml. I'd estimate the readability to be 3 ml, which would be 0.3%. Because of this and the care I took checking with two buckets, I'd estimate my measurement uncertainty levels to be on the order of 0.5%.

Cat litter bucket

We have some 4 gallon cat litter buckets that are made by Ropak. These buckets have a cross

section that is a square with rounded corners. The relevant dimensions in cm are:

$$\begin{aligned} D &= \text{top width} = 23.1 \\ d &= \text{bottom width} = 20.7 \\ h &= 33.0 \\ \delta &= 0.3 \\ \sigma &= D/d - 1 = 0.1159 \\ v_x &= 9 \text{ liters} = 9000 \text{ cm}^3 \\ r &= \text{corner radius} = 3.2 \end{aligned}$$

The area of the bottom A_0 is d^2 with a correction subtracted off for the corner radii; this correction is $(4-\pi)r^2$ where r is the corner radius (see the appendix for derivation). Thus we have $A_0 = 20.7^2 - (4-\pi)3.2^2 = 419.7 \text{ cm}^2$. Similarly, we get for the area of the top of the bucket $A_1 = 23.1^2 - (4-\pi)3.2^2 = 524.8 \text{ cm}^2$. Note: I used the same corner radius for top and bottom of the bucket, but it's possible the radii changed over that distance (I didn't measure them closely). Using equation (1), we get the volume for the whole bucket as 15550 cm^3 or 15.55 liters (4.11 gallons).

Putting these numbers into equation (5) yields

$$x = \frac{\sqrt[3]{1 + \frac{3(0.1159)(9000)}{419.8(33.0)}} - 1}{0.1159} = 0.6060$$

This x is the fraction of the overall bucket height that the level of fluid will assume in the bucket measured vertically from the bottom of the bucket.

Solving for y gives

$$y = (0.6060(33) + 0.3) \sqrt{1 + \left(\frac{0.1159(23.1)}{2(33)} \right)^2} = 20.32 \text{ cm}$$

I used a 1000 ml graduated cylinder and filled the bucket with 9 liters of water. A tape measure showed that the water level was at 20.1 cm measured from the base of the bucket or 2 mm from the calculated value. This demonstrates the calibration accuracy is on the order of a percent or two.

A similar check for 5 liters of added water gave a measured height of $y = 11.8 \text{ cm}$. The calculated value for this situation gave $y = 11.76 \text{ cm}$.

Other applications

The formula is useful for calibrating the volume of things that don't look like the typical bucket. For example, consider a classically-shaped wheelbarrow. If the bottom and top are similar figures and the sides are planes, then you can calculate the volume calibration marks for this wheelbarrow. If the assumption of a frustum isn't valid, then you can use a calibrated bucket to fill the wheelbarrow and make the required volume marks.

You could calculate the volumes of the following wooden planter boxes:



Since these appear to be octagons, it's simple to get the relevant dimensions and areas needed to perform the calculations. The calculations would work for the following planter box:



This box could also be rectangular rather than square -- then the results would depend on which side you wanted to calibrate.

You could even use the formula to calculate the volume of e.g. an overflow pond, as long as the sides sloped in such a way to satisfy the assumptions stated at the beginning.

Python script

I've included a python script called `bucket.py`. It can make these calibration calculations for you (it's what I use to calibrate my buckets). If you run the script, you'll be prompted for the bucket dimensions; otherwise, you can put the bucket's dimensions in a datafile. Run the script with the `-h` option to see a usage statement.

Here's an example datafile:

```
# Test datafile for the bucket.py script.
shape = "round"
title = "Rheem 5 gallon bucket test case"
input_units = "inches"
D = 11.34
d = 10.42
offset = "11.9 mm"
h = 14
```

```
Report("liters", "mm", 1)
```

When the `bucket.py` script is run with the command `python bucket.py bucket.rheem`, the results output are:

```
Rheem 5 gallon bucket test case
```

```
Expected values:
```

```
5 L : 100.9 mm
```

```
10 L : 186.1 mm
```

```
15 L : 268.1 mm
```

```
Rheem 5 gallon bucket test case
```

```
Bucket volume calibration (Tue May 5 10:24:12 2015)
```

```
Default units = inches
```

```
Bucket shape = round
```

```
D = 11.34 inches
```

```
d = 10.42 inches
```

```
offset = 11.9 mm
```

```
h = 14 inches
```

```
Volume:
```

```
21.34 liters
```

```
5.638 gallons
```

```
0.7537 ft3
```

```
1302 in3
```

```
Calibration table (volume in liters, length in mm)
```

Volume	Length
-----	-----
1.	30.01
2.	47.96
3.	65.75
4.	83.38
5.	100.9
6.	118.2
7.	135.4
8.	152.5
9.	169.4
10.	186.1
11.	202.8
12.	219.3
13.	235.7
14.	252.0
15.	268.1
16.	284.1
17.	300.0
18.	315.8
19.	331.5
20.	347.0
21.	362.5

This particular example was used as a test case because I used a graduated cylinder to fill the bucket to a known volume, then measured the slant height with a tape measure. The accuracy was on the order of about 2%.

The calculations assume the bucket is shaped geometrically perfectly; this is probably not the case for most real buckets.

Percentages

The script allows you to enter % for the output units. This causes the output "volume" to be a percentage of a fraction of the bucket's total volume. You'll be prompted for this fraction, which must be in the interval (0, 1].

May 2025 update: uncertainties

This version includes the ability to include uncertainties in the measured values and python's uncertainties library propagates the uncertainty into the measured calibration lengths. This is a useful feature because it gives you a feel for the accuracy of the bucket's calibration marks.

The `bucket.simplot` file gives an example of measurements with uncertainty:

```
# Test datafile for the bucket.py script.
shape = "round"
title = "Simplot 30 lb bucket for strawberries"
input_units = "mm"
D = 270(2)
d = 247.5 ± 1
offset = 7 ± 1
h = 285.(1)
digits = 4
output_volume_units = "l"
output_length_units = "mm"
volume_steps = 1
```

The lines highlighted in red will be discussed below. The script produces the output

```
<string>:1: SyntaxWarning: 'int' object is not callable; perhaps you missed a
comma?
<string>:1: SyntaxWarning: 'int' object is not callable; perhaps you missed a
comma?
Simplot 30 lb bucket for strawberries
```

```
Bucket volume calibration (Tue Nov 18 08:05:49 2025)
```

```
Datafile = bucket.simplot
Default units = mm
Bucket shape = round
D = 270(2) mm
d = 248(1) mm
h = 285(1) mm
offset = 7(1) mm
digits = 4
Volume:
    15.0(1) liters
    3.96(4) gallons
    0.530(5) ft3
    915(9) in3
```

```
Calibration table (volume in l, length in mm)
      Volume      Length
```

1	28(1)
2	48(1)
3	68(1)
4	88(1)
5	108(1)
6	127(1)
7	146(1)
8	165(1)
9	184(2)
10	203(2)
11	221(2)
12	239(2)
13	257(2)
14	275(2)

The uncertainties in the measured values were my type B estimates of how close I was able to measure things. For the diameters, this number includes an estimate of the out-of-roundness of the bucket (plastic buckets are always out of round).

The python uncertainties library uses standard linear uncertainty propagation and assumes no correlation between the measured values (i.e., they are statistically independent). This is probably a pretty good assumption for most buckets unless you know that the typical roughly elliptical shapes e.g. have their major diameters in the same plane.

The lines in red in the above listing use the standard short-form notation for uncertainties. `270(2)` means the standard uncertainty is 2 in the last significant figure of the significand. Here, this means the expression could alternatively be written `270±2`. If you prefer the latter form, use the `-s` option when calling the script. These lines in red caused the above warnings to stderr in red from the python interpreter (`SyntaxWarning` stuff above). The interpreter sees `D = 270(2)` in an expression and thinks it's an integer expression trying to be called as a function.

If you're interested in the code details, in `ReadData()`, the line in the data file is evaluated using an `exec()` call, but the string value `"D = 270(2)"` for `_line` results in an exception, reported to stderr by the python interpreter. This exception is caught and `InterpretUncertainty()` is called to evaluate the uncertainty expression.

There are various ways to ignore such warnings; you can find them by a web search.

When you get these warning messages, you can use the `-D` option to the script to cause the datafile line being interpreted to be printed out in color; this lets you see the warning message just after the offending line in the datafile.

Appendix: Derivation of formula

We assume the relevant volume is a frustum of a generalized cone with arbitrary cross section. For a [frustum](#) with height h that has areas A_0 and A_1 on the ends, the volume of the frustum is (see [here](#) for a derivation):

$$V = \frac{h}{3}(A_0 + A_1 + \sqrt{A_0 A_1}) \quad (1)$$

Note this formula was known nearly 4000 years ago.

A test to assure yourself that you're dealing with a frustum is to extend the sides. If they meet in a point, then you have a frustum.

The problem we want to solve is that we pick some value of volume we want, for example, an even number of gallons. Then we solve formula (1) for the height h needed to give the desired volume. It

turns out this results in a cubic equation in the unknown height and this equation has one real root (A_1 is a function of h).

Note it has to be a frustum -- this lets the volumes of the intermediate frustums vary proportionately between zero and the volume V . If the volume is not a frustum, you'll have to do some integration in general to find its volume.

We'll suppose that the linear dimensions of the planar figure making up area A_1 (the top of the bucket) are $1+\sigma$ times those making up area A_0 where $\sigma \geq 0$ (this gives a conventional upwards-opening bucket¹). We have $A_1 = (1+\sigma)^2 A_0$. We also require that the planar figure that is the top of the bucket is similar in a geometrical sense to the planar figure of the bottom of the bucket and with no rotation about an axis perpendicular to the plane. Here, similar means that one figure can be uniformly scaled and translated in such a way that it can be transformed into the other figure.

In cross section, the bucket looks like

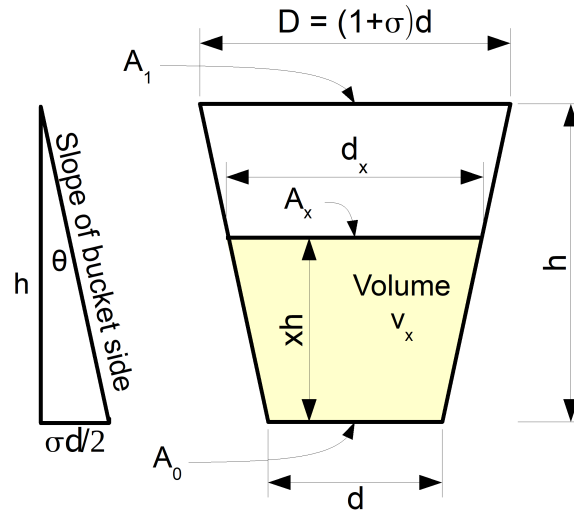


Figure 3

where the dimension d characterizes the bottom. The top is the same planar shape scaled up in size by the factor $1+\sigma$; we'll set $D = d(1+\sigma)$. We'll assume that the planar areas A_0 and A_1 are perpendicular to the local gravitational force vector, meaning that a surface of water at rest in the bucket is parallel to these two surfaces.

The cross-sectional² shape of the bucket can be any closed planar figure with a well-defined area -- a circle, ellipse, polygon, etc.

Suppose we wish to determine the fraction $x \in [0,1]$ of the height h such that the volume of the frustum is v_x (the colored area in the above picture) where $v_x \leq V$. Let A_x be the area of the top of the colored volume at height xh . The relevant linear dimension for the top of the colored volume is $d_x = d(1+x\sigma)$. Then $A_x = (1+x\sigma)^2 A_0$. Substituting into equation (1) gives the volume of the colored area as

$$v_x = \frac{xh}{3} \left(A_0 + (1+x\sigma)^2 A_0 + \sqrt{A_0 A_0 (1+x\sigma)^2} \right)$$

This results in a cubic equation to be solved for x :

$$\frac{\sigma^2}{3} x^3 + \sigma x^2 + x - \frac{v_x}{h A_0} = 0$$

1 And helps make the problem algebraically more tractable, although it rules out degenerate cases like a cone where A_0 is zero.

2 The plane of the cross-section is perpendicular to the gravitational acceleration vector.

This equation can be put into Mathematica and solved for x (s is σ):

```
f = s^2/3*x^3 + s*x^2 + x - vx/(h*A0);  
Simplify[Solve[f == 0, x]]
```

which yields the one real root

$$-1 + \frac{(A_0 h + 3 s v_x)^{1/3}}{A_0^{1/3} h^{1/3} s}$$

Alternatively, if you have [sympy](#), you can use the following script to get the solution:

```
from sympy.solvers import solve
from sympy import Symbol
x, s, vx, h, A0 = map(Symbol, ("x", "s", "vx", "h", "A0"))
print solve(s**2/3*x**3 + s*x**2 + x - vx/(h*A0), x)
```

This produces the real root

$$-(-1/s**3 - 3*vx/(A0*h*s**2))**(1/3) - 1/s$$

which is the same result.

Finally, you can enter `solve s^2/3*x^3 + s*x^2 + x - a = 0 for x` into [Wolfram Alpha](#) and you'll see similar results.

The solution is

$$x = \frac{1}{\sigma} \left[\sqrt[3]{1 + \frac{3\sigma v_x}{A_0 h}} - 1 \right] \quad (2)$$

As σ approaches zero³ (i.e., the bucket shape approaches a generalized cylinder), x approaches $\frac{v_x}{A_0 h}$; this is the fraction of the total volume of the vertical cylinder from the base, as we'd expect. xh is the vertical height from the inside bottom of the bucket that marks the volume v ; it is the height of the liquid surface of volume v_x measured along the gravitational vector from the bottom of the bucket. A real bucket will have its inside bottom a distance δ (what we called the offset in Figure 2 above) from the plane surface that the bucket sits on. Thus, the vertical mark will be at a vertical height of $xh + \delta$.

Plastic buckets are tapered so the bucket can be withdrawn from its mold. To allow us to measure along the side of the bucket, we need to increase $xh + \delta$ by the factor $1/\cos\theta$ where θ is the angle the side of the bucket makes with the gravitational vector. Thus, the mark measured along the bucket's side to yield the volume v_x is

$$y = \frac{xh + \delta}{\cos\theta} \quad (3)$$

where from similar triangles

$$\cos\theta = \left[1 + \left(\frac{\sigma d_x}{2h} \right)^2 \right]^{-1/2} \quad (4)$$

(or Figure 3 shows $\tan\theta = \frac{\sigma d}{2h}$) and d_x is the relevant linear dimension of the top that is cut by the plane of the cross section (e.g., diameter, width) where the volume marks will be made. This particular d_x , which is the width of a square bucket or the diameter of a round bucket, both at the bucket top, is the only place this particular dimension is needed -- and it's only used to calculate the angle of the bucket's side with respect to the vertical. If you were marking e.g. a bucket that was

3 You'll have to use l'Hopital's rule to calculate the limit or e.g. let Mathematica do it for you.

made with a polygonal cross section, the relevant value of d_x would be the distance across the top where a plane parallel to the bucket's longitudinal axis contains the axis and also passes through the side surface where you want to make the volume calibration marks. If you make a sketch, you'll see that this cosine was gotten from similar triangles and the Pythagorean theorem. Here's a picture that shows how the required d_x is gotten; the cutting plane is perpendicular to the top and bottom surfaces (i.e., this cutting plane fully contains a gravitational vector).

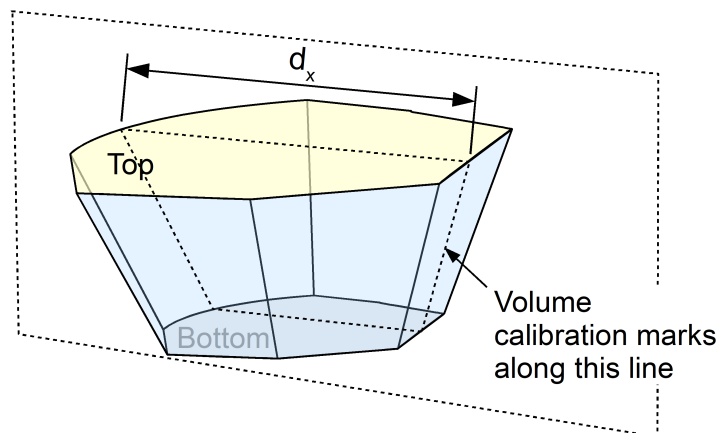


Figure 4

The angle of the side of interest is the angle with respect to the gravitational vector; if the bucket isn't symmetrical, then use a d_x that is appropriate for that particular side. Note, for example, that the centers of symmetry of the two planes don't have to lie on the same gravitational vector line. In other words, the two closed areas A_0 and A_1 can be moved around in the planes that contain them and the volume of the relevant solid doesn't change (shown by equation (1) above) -- but the angle along the bucket edge does and varies solely because of the angle of the side with respect to the gravitational vector.

This yields the final equation

$$\boxed{\begin{aligned} y &= (xh + \delta) \sqrt{1 + \left(\frac{\sigma(1+x)d}{2h} \right)^2} \\ \text{where } x &= \frac{1}{\sigma} \left[\sqrt[3]{1 + \frac{3\sigma v_x}{A_0 h}} - 1 \right] \end{aligned}} \quad (5)$$

Make sure you measure all linear dimensions in the same units (and the areas must be in the square of these units). Don't miss the cube root in equation (5).

While x could be substituted into the expression for y and simplified a bit, it's useful to see the value for x , which is the fraction of the overall height of the bucket that represents the volume v_x . If you get a value for x such that $0 < x \leq 1$ isn't true, then you know there's a mistake in the calculation somewhere.

This formula gives the slant distance **y** (shown in red in Figure 5) along the outside of the bucket and measured from the bottom of the bucket such that the top of the planar surface A is located so that the shaded volume of the frustum is v_x . **This distance y lets you calibrate the bucket with a tape measure.**

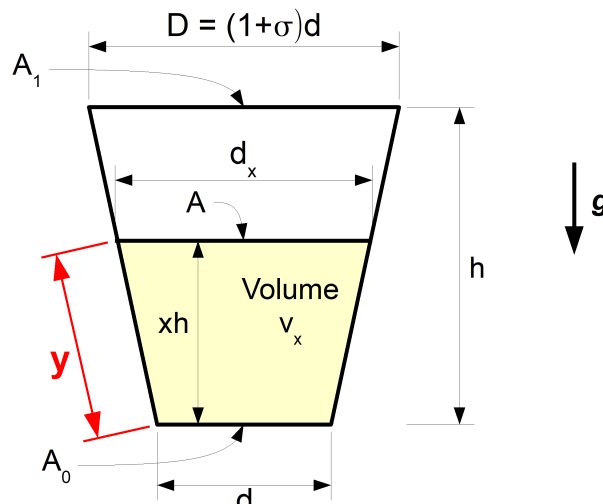


Figure 5

The symbols in the figure and equation (5) are ($\mathbf{g}$ is the gravitational vector):

- σ = linear dimension characterizing how much larger the top area A_1 is from the bottom area A_0
- d = linear dimension characterizing the area A_0 of the bottom of the frustum
- d_x = linear dimension characterizing the area A of the top of the colored frustum
- δ = offset of bucket (see Figure 2 on page 2)
- A_0 = bottom planar area of frustum (is in a plane perpendicular to $\mathbf{g}$)
- A_1 = top planar area of frustum (is in a plane perpendicular to $\mathbf{g}$)
- A = area of top of colored frustum at height xh (is in a plane perpendicular to $\mathbf{g}$)
- h = height of frustum
- x = fraction of height h that has the planar area A such that the colored volume is v_x
- v_x = volume of the colored frustum in Figure 5
- y = distance to measure along edge of bucket to mark volume v_x**

Bucket calibration algorithm

Because the formulas and symbols can be confusing (they were confusing to me when I came back and read this material again), here's how to use equation (5) to calibrate a round bucket as shown in Figure 2; modify the instructions appropriately if the cross-section of the frustum is not circular.

1. For the bucket, measure the dimensions d , D , h , and offset = δ as shown in Figure 2.
2. Calculate $\sigma = \frac{D}{d} - 1$.
3. Calculate $A_0 = \pi \frac{d^2}{4}$. You'll have to use a different formula for the area if the cross section of the bucket is not round.
4. Pick the volume v_x that you want to mark on the side of the bucket.
5. Calculate x in equation (5) (don't miss the cube root), then calculate the value of y (the slant distance up the side of the bucket that marks volume v_x). Mark off the distance y on the side of the bucket.

Here's an example calculation for the Rheem bucket with a round cross section. The dimensions in inches were

$$\begin{aligned} D &= 11.34 \\ d &= 10.42 \\ h &= 14.00 \end{aligned}$$

$$\text{offset} = \delta = 0.47$$

We calculate $\sigma = \frac{11.34}{10.42} - 1 = 0.0882917$. The area A_0 of the bottom of the bucket is 85.2757 in^3 .

Suppose we want to mark the side of the bucket where the volume v_x is 10 liters or 610.237 in^3 . Plugging these numbers into equation (5), we calculate

$$x = 0.4896708$$

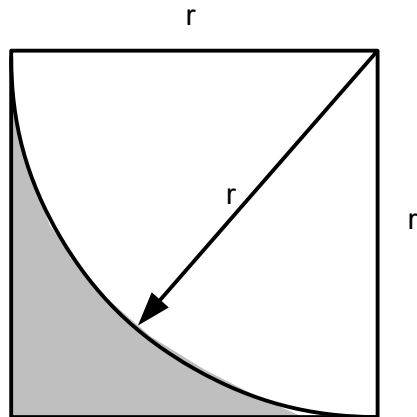
$$y = 7.334$$

The calculated slant distance up the side of the bucket to mark 10 liters is 7.334 inches or 186.3 mm. I checked the calibration by filling this bucket with 10 liters from a Nalgene 1000 ml graduated cylinder. The measured distance of the water level up the side of the bucket was 186.5 mm.

I recommend using the python script to print out calibration tables, as you will save time over doing the calculations manually.

Bucket with rounded corners

To calculate the area of a rectangular or square bucket with rounded corners, you need to subtract off a correction from the rectangle's area. Here's a picture:



We want the area of the gray portion (sometimes called a spandrel). The square's area is r^2 . The circular part is one-fourth the area of a circle, so we subtract off $\frac{\pi}{4} r^2$. Since there are four corners, the area of the four gray portions is $(4 - \pi)r^2$. This is the correction that must be subtracted from a rectangular area to account for the curved corners.